

Annex 1 - Estudi de la complexitat:

Resultats de les execucions

Els fitxers de prova es troben a la carpeta `/jocs-prova`.

Bàsic

Pel bàsic només hem tingut en compte el temps d'execució de problemes que es poden resoldre (perquè els que no tenen solució tarden massa en resoldre's, veure cas mida 14).

Shell

```
python3 ./generate_problems.py 3 --level basic
./Metric-FF/ff -O -o ./domain-basic.pddl -f ./jocs-prova/problem-basic-auto-3.pddl
```

Mida 3

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R3 H2
         1: ASSIGNAR R1 H1
         2: ASSIGNAR R2 H1
```

```
time spent:    0.00 seconds instantiating 0 easy, 4 hard action templates
               0.00 seconds reachability analysis, yielding 77 facts and 4 actions
               0.00 seconds creating final representation with 24 relevant facts, 0
relevant fluents
               0.00 seconds computing LNF
               0.00 seconds building connectivity graph
               0.00 seconds searching, evaluating 8 states, to a max depth of 0
               0.00 seconds total time
```

Mida 5

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R5 H3
         1: ASSIGNAR R2 H3
         2: ASSIGNAR R1 H3
         3: ASSIGNAR R3 H1
         4: ASSIGNAR R4 H1
```

```
time spent:    0.00 seconds instantiating 0 easy, 11 hard action templates
              0.00 seconds reachability analysis, yielding 132 facts and 11 actions
              0.00 seconds creating final representation with 63 relevant facts, 0
relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.00 seconds searching, evaluating 28 states, to a max depth of 0
              0.00 seconds total time
```

Mida 7

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R7 H4
         1: ASSIGNAR R6 H4
         2: ASSIGNAR R2 H4
         3: ASSIGNAR R1 H4
         4: ASSIGNAR R3 H3
         5: ASSIGNAR R4 H3
         6: ASSIGNAR R5 H1
```

```
time spent:    0.00 seconds instantiating 0 easy, 20 hard action templates
              0.00 seconds reachability analysis, yielding 190 facts and 20 actions
              0.00 seconds creating final representation with 106 relevant facts, 0
relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.00 seconds searching, evaluating 74 states, to a max depth of 0
              0.00 seconds total time
```

Mida 9

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R2 H5
         1: ASSIGNAR R4 H5
         2: ASSIGNAR R1 H5
         3: ASSIGNAR R8 H5
         4: ASSIGNAR R7 H5
         5: ASSIGNAR R5 H5
         6: ASSIGNAR R6 H4
         7: ASSIGNAR R9 H1
         8: ASSIGNAR R3 H1
```

```
time spent:    0.00 seconds instantiating 0 easy, 29 hard action templates
              0.00 seconds reachability analysis, yielding 266 facts and 29 actions
              0.00 seconds creating final representation with 185 relevant facts, 0
relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.00 seconds searching, evaluating 125 states, to a max depth of 0
              0.00 seconds total time
```

Mida 10

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R8 H5
         1: ASSIGNAR R4 H5
         2: ASSIGNAR R7 H5
         3: ASSIGNAR R5 H5
         4: ASSIGNAR R9 H5
         5: ASSIGNAR R2 H3
         6: ASSIGNAR R10 H1
         7: ASSIGNAR R1 H1
         8: ASSIGNAR R6 H1
         9: ASSIGNAR R3 H1
```

```
time spent:    0.00 seconds instantiating 0 easy, 24 hard action templates
              0.00 seconds reachability analysis, yielding 255 facts and 24 actions
              0.00 seconds creating final representation with 166 relevant facts, 0
relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.00 seconds searching, evaluating 108 states, to a max depth of 0
              0.00 seconds total time
```

Mida 11

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R4 H6
         1: ASSIGNAR R6 H6
         2: ASSIGNAR R5 H6
         3: ASSIGNAR R9 H6
         4: ASSIGNAR R1 H5
         5: ASSIGNAR R2 H5
```

```
6: ASSIGNAR R7 H5
7: ASSIGNAR R8 H5
8: ASSIGNAR R11 H3
9: ASSIGNAR R10 H3
10: ASSIGNAR R3 H1
```

```
time spent: 0.00 seconds instantiating 0 easy, 51 hard action templates
            0.00 seconds reachability analysis, yielding 332 facts and 51 actions
            0.00 seconds creating final representation with 231 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.01 seconds searching, evaluating 261 states, to a max depth of 0
            0.01 seconds total time
```

Mida 12

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R9 H6
      1: ASSIGNAR R3 H6
      2: ASSIGNAR R8 H6
      3: ASSIGNAR R1 H6
      4: ASSIGNAR R12 H5
      5: ASSIGNAR R6 H4
      6: ASSIGNAR R10 H4
      7: ASSIGNAR R4 H4
      8: ASSIGNAR R2 H4
      9: ASSIGNAR R7 H4
     10: ASSIGNAR R11 H3
     11: ASSIGNAR R5 H3
```

```
time spent: 0.00 seconds instantiating 0 easy, 58 hard action templates
            0.00 seconds reachability analysis, yielding 365 facts and 58 actions
            0.00 seconds creating final representation with 288 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.01 seconds searching, evaluating 330 states, to a max depth of 0
            0.01 seconds total time
```

Mida 13

Shell

ff: found legal plan as follows

```

step    0: ASSIGNAR R3 H7
        1: ASSIGNAR R7 H7
        2: ASSIGNAR R10 H6
        3: ASSIGNAR R6 H5
        4: ASSIGNAR R9 H5
        5: ASSIGNAR R12 H5
        6: ASSIGNAR R11 H3
        7: ASSIGNAR R8 H3
        8: ASSIGNAR R4 H3
        9: ASSIGNAR R13 H3
       10: ASSIGNAR R5 H3
       11: ASSIGNAR R2 H3
       12: ASSIGNAR R1 H2

time spent:  0.00 seconds instantiating 0 easy, 56 hard action templates
            0.00 seconds reachability analysis, yielding 390 facts and 56 actions
            0.00 seconds creating final representation with 278 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.00 seconds searching, evaluating 319 states, to a max depth of 0
            0.00 seconds total time

```

Mida 14 (cas sense solució)

```

Shell
ff: search configuration is  best-first on 1*g(s) + 5*h(s) where
    metric is  plan length

advancing to distance:  14
                      13
                      12
                      11
                      10
                      9
                      8
                      7
                      6

best first search space empty! problem proven unsolvable.

time spent:  0.00 seconds instantiating 0 easy, 46 hard action templates
            0.00 seconds reachability analysis, yielding 383 facts and 46 actions
            0.00 seconds creating final representation with 272 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph

```

```
88.77 seconds searching, evaluating 6190976 states, to a max depth of 0
88.77 seconds total time
```

Mida 14 (amb solució)

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R7 H7
         1: ASSIGNAR R3 H7
         2: ASSIGNAR R6 H7
         3: ASSIGNAR R11 H7
         4: ASSIGNAR R1 H7
         5: ASSIGNAR R2 H6
         6: ASSIGNAR R12 H6
         7: ASSIGNAR R8 H6
         8: ASSIGNAR R5 H6
         9: ASSIGNAR R4 H5
        10: ASSIGNAR R10 H5
        11: ASSIGNAR R9 H5
        12: ASSIGNAR R14 H4
        13: ASSIGNAR R13 H4
```

```
time spent:  0.00 seconds instantiating 0 easy, 81 hard action templates
            0.00 seconds reachability analysis, yielding 448 facts and 81 actions
            0.00 seconds creating final representation with 367 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.02 seconds searching, evaluating 527 states, to a max depth of 0
            0.02 seconds total time
```

Mida 15

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R10 H8
         1: ASSIGNAR R12 H8
         2: ASSIGNAR R4 H8
         3: ASSIGNAR R2 H8
         4: ASSIGNAR R6 H8
         5: ASSIGNAR R5 H8
         6: ASSIGNAR R8 H8
         7: ASSIGNAR R14 H8
         8: ASSIGNAR R9 H7
         9: ASSIGNAR R7 H7
```

```
10: ASSIGNAR R15 H7
11: ASSIGNAR R3 H6
12: ASSIGNAR R13 H6
13: ASSIGNAR R1 H5
14: ASSIGNAR R11 H4
```

```
time spent:  0.00 seconds instantiating 0 easy, 96 hard action templates
            0.00 seconds reachability analysis, yielding 545 facts and 96 actions
            0.00 seconds creating final representation with 484 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.02 seconds searching, evaluating 695 states, to a max depth of 0
            0.02 seconds total time
```

Mida 17

Shell

ff: found legal plan as follows

```
step  0: ASSIGNAR R3 H9
      1: ASSIGNAR R6 H9
      2: ASSIGNAR R14 H9
      3: ASSIGNAR R13 H9
      4: ASSIGNAR R10 H9
      5: ASSIGNAR R1 H9
      6: ASSIGNAR R5 H9
      7: ASSIGNAR R12 H8
      8: ASSIGNAR R9 H8
      9: ASSIGNAR R15 H8
     10: ASSIGNAR R16 H8
     11: ASSIGNAR R11 H8
     12: ASSIGNAR R2 H7
     13: ASSIGNAR R4 H7
     14: ASSIGNAR R8 H3
     15: ASSIGNAR R17 H3
     16: ASSIGNAR R7 H2
```

```
time spent:  0.00 seconds instantiating 0 easy, 112 hard action templates
            0.00 seconds reachability analysis, yielding 613 facts and 112 actions
            0.00 seconds creating final representation with 540 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.02 seconds searching, evaluating 893 states, to a max depth of 0
            0.02 seconds total time
```

Mida 19

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R17 H10
        1: ASSIGNAR R1  H10
        2: ASSIGNAR R4  H10
        3: ASSIGNAR R16 H10
        4: ASSIGNAR R5  H10
        5: ASSIGNAR R15 H9
        6: ASSIGNAR R12 H9
        7: ASSIGNAR R13 H9
        8: ASSIGNAR R9  H9
        9: ASSIGNAR R3  H9
       10: ASSIGNAR R8  H8
       11: ASSIGNAR R14 H8
       12: ASSIGNAR R7  H8
       13: ASSIGNAR R2  H8
       14: ASSIGNAR R10 H6
       15: ASSIGNAR R11 H6
       16: ASSIGNAR R6  H6
       17: ASSIGNAR R19 H4
       18: ASSIGNAR R18 H4
```

```
time spent:  0.00 seconds instantiating 0 easy, 151 hard action templates
            0.00 seconds reachability analysis, yielding 730 facts and 151 actions
            0.00 seconds creating final representation with 671 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.03 seconds searching, evaluating 1318 states, to a max depth of 0
            0.03 seconds total time
```

Mida 20

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R20 H10
        1: ASSIGNAR R1  H10
        2: ASSIGNAR R6  H10
        3: ASSIGNAR R8  H10
        4: ASSIGNAR R3  H10
        5: ASSIGNAR R2  H10
        6: ASSIGNAR R12 H10
        7: ASSIGNAR R13 H10
        8: ASSIGNAR R5  H10
        9: ASSIGNAR R7  H9
       10: ASSIGNAR R9  H9
       11: ASSIGNAR R19 H9
```



```
12: ASSIGNAR R4 H9
13: ASSIGNAR R10 H9
14: ASSIGNAR R18 H8
15: ASSIGNAR R11 H8
16: ASSIGNAR R16 H8
17: ASSIGNAR R15 H8
18: ASSIGNAR R17 H5
19: ASSIGNAR R14 H5
```

```
time spent:  0.00 seconds instantiating 0 easy, 161 hard action templates
            0.00 seconds reachability analysis, yielding 709 facts and 161 actions
            0.00 seconds creating final representation with 617 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.03 seconds searching, evaluating 1531 states, to a max depth of 0
            0.03 seconds total time
```

Mida 22

Shell

ff: found legal plan as follows

```
step  0: ASSIGNAR R4 H11
      1: ASSIGNAR R3 H11
      2: ASSIGNAR R20 H11
      3: ASSIGNAR R17 H11
      4: ASSIGNAR R22 H11
      5: ASSIGNAR R9 H11
      6: ASSIGNAR R2 H10
      7: ASSIGNAR R16 H10
      8: ASSIGNAR R7 H10
      9: ASSIGNAR R1 H10
     10: ASSIGNAR R15 H10
     11: ASSIGNAR R8 H10
     12: ASSIGNAR R18 H9
     13: ASSIGNAR R13 H9
     14: ASSIGNAR R12 H9
     15: ASSIGNAR R19 H7
     16: ASSIGNAR R21 H7
     17: ASSIGNAR R5 H7
     18: ASSIGNAR R10 H7
     19: ASSIGNAR R6 H7
     20: ASSIGNAR R11 H6
     21: ASSIGNAR R14 H6
```

```
time spent:  0.00 seconds instantiating 0 easy, 186 hard action templates
            0.00 seconds reachability analysis, yielding 824 facts and 186 actions
```

```
0.00 seconds creating final representation with 758 relevant facts, 0
relevant fluents
0.00 seconds computing LNF
0.00 seconds building connectivity graph
0.04 seconds searching, evaluating 1817 states, to a max depth of 0
0.04 seconds total time
```

Mida 24

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R2 H12
1: ASSIGNAR R5 H12
2: ASSIGNAR R14 H12
3: ASSIGNAR R7 H12
4: ASSIGNAR R6 H12
5: ASSIGNAR R11 H12
6: ASSIGNAR R12 H12
7: ASSIGNAR R20 H12
8: ASSIGNAR R17 H11
9: ASSIGNAR R10 H11
10: ASSIGNAR R16 H11
11: ASSIGNAR R3 H11
12: ASSIGNAR R23 H11
13: ASSIGNAR R1 H11
14: ASSIGNAR R21 H11
15: ASSIGNAR R24 H10
16: ASSIGNAR R4 H10
17: ASSIGNAR R13 H9
18: ASSIGNAR R19 H9
19: ASSIGNAR R18 H8
20: ASSIGNAR R15 H7
21: ASSIGNAR R8 H7
22: ASSIGNAR R22 H7
23: ASSIGNAR R9 H4
```

```
time spent: 0.00 seconds instantiating 0 easy, 232 hard action templates
0.00 seconds reachability analysis, yielding 964 facts and 232 actions
0.00 seconds creating final representation with 928 relevant facts, 0
relevant fluents
0.00 seconds computing LNF
0.00 seconds building connectivity graph
0.07 seconds searching, evaluating 2565 states, to a max depth of 0
0.07 seconds total time
```

Mida 26

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R20 H13
        1: ASSIGNAR R15 H13
        2: ASSIGNAR R6 H13
        3: ASSIGNAR R22 H13
        4: ASSIGNAR R26 H13
        5: ASSIGNAR R14 H13
        6: ASSIGNAR R11 H13
        7: ASSIGNAR R7 H13
        8: ASSIGNAR R1 H13
        9: ASSIGNAR R12 H12
       10: ASSIGNAR R4 H12
       11: ASSIGNAR R3 H12
       12: ASSIGNAR R24 H12
       13: ASSIGNAR R10 H12
       14: ASSIGNAR R13 H11
       15: ASSIGNAR R23 H8
       16: ASSIGNAR R8 H8
       17: ASSIGNAR R25 H8
       18: ASSIGNAR R5 H8
       19: ASSIGNAR R21 H8
       20: ASSIGNAR R19 H6
       21: ASSIGNAR R2 H6
       22: ASSIGNAR R16 H6
       23: ASSIGNAR R9 H5
       24: ASSIGNAR R17 H5
       25: ASSIGNAR R18 H5
```

```
time spent:  0.00 seconds instantiating 0 easy, 179 hard action templates
            0.00 seconds reachability analysis, yielding 897 facts and 179 actions
            0.00 seconds creating final representation with 783 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.07 seconds searching, evaluating 2013 states, to a max depth of 0
            0.07 seconds total time
```

Mida 28

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R19 H14
        1: ASSIGNAR R28 H14
        2: ASSIGNAR R20 H14
        3: ASSIGNAR R7 H14
        4: ASSIGNAR R17 H14
        5: ASSIGNAR R14 H14
        6: ASSIGNAR R23 H14
```

```
7: ASSIGNAR R24 H13
8: ASSIGNAR R13 H13
9: ASSIGNAR R26 H13
10: ASSIGNAR R15 H13
11: ASSIGNAR R4 H13
12: ASSIGNAR R22 H13
13: ASSIGNAR R18 H12
14: ASSIGNAR R21 H12
15: ASSIGNAR R8 H12
16: ASSIGNAR R1 H12
17: ASSIGNAR R6 H11
18: ASSIGNAR R11 H11
19: ASSIGNAR R25 H11
20: ASSIGNAR R9 H10
21: ASSIGNAR R3 H9
22: ASSIGNAR R12 H9
23: ASSIGNAR R10 H9
24: ASSIGNAR R27 H9
25: ASSIGNAR R2 H4
26: ASSIGNAR R16 H4
27: ASSIGNAR R5 H2
```

```
time spent: 0.01 seconds instantiating 0 easy, 272 hard action templates
            0.00 seconds reachability analysis, yielding 1111 facts and 272 actions
            0.00 seconds creating final representation with 1054 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.25 seconds searching, evaluating 3051 states, to a max depth of 0
            0.26 seconds total time
```

Mida 30

Shell

Mida 30

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R9 H15
      1: ASSIGNAR R12 H15
      2: ASSIGNAR R14 H15
      3: ASSIGNAR R15 H14
      4: ASSIGNAR R20 H14
      5: ASSIGNAR R22 H14
```

```
6: ASSIGNAR R23 H14
7: ASSIGNAR R25 H14
8: ASSIGNAR R13 H14
9: ASSIGNAR R21 H13
10: ASSIGNAR R8 H13
11: ASSIGNAR R11 H13
12: ASSIGNAR R17 H13
13: ASSIGNAR R27 H13
14: ASSIGNAR R29 H13
15: ASSIGNAR R5 H13
16: ASSIGNAR R1 H13
17: ASSIGNAR R16 H12
18: ASSIGNAR R6 H12
19: ASSIGNAR R24 H12
20: ASSIGNAR R30 H11
21: ASSIGNAR R10 H11
22: ASSIGNAR R18 H10
23: ASSIGNAR R26 H8
24: ASSIGNAR R28 H7
25: ASSIGNAR R3 H6
26: ASSIGNAR R7 H6
27: ASSIGNAR R4 H5
28: ASSIGNAR R19 H4
29: ASSIGNAR R2 H4
```

```
time spent: 0.01 seconds instantiating 0 easy, 342 hard action templates
            0.00 seconds reachability analysis, yielding 1250 facts and 342 actions
            0.00 seconds creating final representation with 1198 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.01 seconds building connectivity graph
            0.17 seconds searching, evaluating 4475 states, to a max depth of 0
            0.19 seconds total time
```

Mida 35

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R1 H18
      1: ASSIGNAR R2 H18
      2: ASSIGNAR R22 H18
      3: ASSIGNAR R16 H18
      4: ASSIGNAR R9 H18
      5: ASSIGNAR R13 H18
      6: ASSIGNAR R3 H18
      7: ASSIGNAR R11 H18
      8: ASSIGNAR R8 H18
      9: ASSIGNAR R23 H17
     10: ASSIGNAR R5 H17
```

```
11: ASSIGNAR R29 H17
12: ASSIGNAR R25 H17
13: ASSIGNAR R34 H17
14: ASSIGNAR R19 H17
15: ASSIGNAR R12 H17
16: ASSIGNAR R20 H17
17: ASSIGNAR R33 H16
18: ASSIGNAR R17 H16
19: ASSIGNAR R27 H16
20: ASSIGNAR R26 H16
21: ASSIGNAR R4 H16
22: ASSIGNAR R18 H16
23: ASSIGNAR R15 H16
24: ASSIGNAR R14 H15
25: ASSIGNAR R30 H15
26: ASSIGNAR R31 H15
27: ASSIGNAR R32 H14
28: ASSIGNAR R35 H13
29: ASSIGNAR R7 H13
30: ASSIGNAR R10 H13
31: ASSIGNAR R28 H13
32: ASSIGNAR R21 H2
33: ASSIGNAR R24 H2
34: ASSIGNAR R6 H2
```

```
time spent: 0.00 seconds instantiating 0 easy, 412 hard action templates
            0.00 seconds reachability analysis, yielding 1470 facts and 412 actions
            0.00 seconds creating final representation with 1378 relevant facts, 0
relevant fluents
            0.01 seconds computing LNF
            0.00 seconds building connectivity graph
            0.33 seconds searching, evaluating 6605 states, to a max depth of 0
            0.34 seconds total time
```

Mida 40

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R15 H20
      1: ASSIGNAR R20 H20
      2: ASSIGNAR R9 H20
      3: ASSIGNAR R4 H20
      4: ASSIGNAR R1 H20
      5: ASSIGNAR R8 H20
      6: ASSIGNAR R36 H20
      7: ASSIGNAR R7 H20
      8: ASSIGNAR R14 H20
      9: ASSIGNAR R6 H20
     10: ASSIGNAR R28 H19
```

```
11: ASSIGNAR R39 H19
12: ASSIGNAR R16 H19
13: ASSIGNAR R27 H19
14: ASSIGNAR R12 H19
15: ASSIGNAR R18 H18
16: ASSIGNAR R21 H18
17: ASSIGNAR R40 H18
18: ASSIGNAR R34 H18
19: ASSIGNAR R24 H18
20: ASSIGNAR R30 H18
21: ASSIGNAR R29 H17
22: ASSIGNAR R23 H17
23: ASSIGNAR R13 H17
24: ASSIGNAR R19 H16
25: ASSIGNAR R35 H16
26: ASSIGNAR R10 H16
27: ASSIGNAR R31 H15
28: ASSIGNAR R33 H15
29: ASSIGNAR R22 H15
30: ASSIGNAR R37 H15
31: ASSIGNAR R26 H14
32: ASSIGNAR R25 H14
33: ASSIGNAR R3 H14
34: ASSIGNAR R11 H12
35: ASSIGNAR R2 H12
36: ASSIGNAR R5 H12
37: ASSIGNAR R32 H12
38: ASSIGNAR R17 H6
39: ASSIGNAR R38 H6
```

```
time spent: 0.00 seconds instantiating 0 easy, 580 hard action templates
            0.01 seconds reachability analysis, yielding 1786 facts and 580 actions
            0.00 seconds creating final representation with 1712 relevant facts, 0
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.53 seconds searching, evaluating 10024 states, to a max depth of 0
            0.54 seconds total time
```

Mida 50

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R2 H25
      1: ASSIGNAR R27 H25
      2: ASSIGNAR R1 H25
      3: ASSIGNAR R15 H25
      4: ASSIGNAR R25 H25
      5: ASSIGNAR R42 H25
```

6: ASSIGNAR R34 H25
7: ASSIGNAR R20 H25
8: ASSIGNAR R21 H25
9: ASSIGNAR R9 H25
10: ASSIGNAR R23 H25
11: ASSIGNAR R7 H24
12: ASSIGNAR R31 H24
13: ASSIGNAR R45 H24
14: ASSIGNAR R29 H24
15: ASSIGNAR R3 H24
16: ASSIGNAR R19 H24
17: ASSIGNAR R5 H24
18: ASSIGNAR R26 H24
19: ASSIGNAR R13 H24
20: ASSIGNAR R28 H23
21: ASSIGNAR R10 H23
22: ASSIGNAR R24 H23
23: ASSIGNAR R43 H23
24: ASSIGNAR R39 H23
25: ASSIGNAR R16 H23
26: ASSIGNAR R49 H22
27: ASSIGNAR R30 H22
28: ASSIGNAR R6 H22
29: ASSIGNAR R35 H21
30: ASSIGNAR R12 H21
31: ASSIGNAR R36 H21
32: ASSIGNAR R33 H21
33: ASSIGNAR R14 H21
34: ASSIGNAR R46 H21
35: ASSIGNAR R8 H21
36: ASSIGNAR R41 H20
37: ASSIGNAR R17 H20
38: ASSIGNAR R4 H20
39: ASSIGNAR R38 H20
40: ASSIGNAR R40 H20
41: ASSIGNAR R44 H19
42: ASSIGNAR R18 H19
43: ASSIGNAR R47 H19
44: ASSIGNAR R37 H18
45: ASSIGNAR R22 H18
46: ASSIGNAR R11 H18
47: ASSIGNAR R48 H17
48: ASSIGNAR R32 H17
49: ASSIGNAR R50 H16

time spent: 0.01 seconds instantiating 0 easy, 982 hard action templates
0.00 seconds reachability analysis, yielding 2537 facts and 982 actions
0.00 seconds creating final representation with 2492 relevant facts, 0
relevant fluents
0.03 seconds computing LNF
0.00 seconds building connectivity graph
1.98 seconds searching, evaluating 22318 states, to a max depth of 0
2.02 seconds total time

Mida 60

Shell

ff: found legal plan as follows

```
step  0: ASSIGNAR R23 H30
      1: ASSIGNAR R3  H30
      2: ASSIGNAR R8  H30
      3: ASSIGNAR R21 H30
      4: ASSIGNAR R54 H30
      5: ASSIGNAR R29 H30
      6: ASSIGNAR R18 H30
      7: ASSIGNAR R20 H30
      8: ASSIGNAR R24 H30
      9: ASSIGNAR R6  H29
     10: ASSIGNAR R15 H29
     11: ASSIGNAR R1  H29
     12: ASSIGNAR R14 H29
     13: ASSIGNAR R2  H29
     14: ASSIGNAR R5  H29
     15: ASSIGNAR R45 H29
     16: ASSIGNAR R41 H28
     17: ASSIGNAR R44 H28
     18: ASSIGNAR R4  H28
     19: ASSIGNAR R37 H28
     20: ASSIGNAR R43 H27
     21: ASSIGNAR R16 H27
     22: ASSIGNAR R30 H27
     23: ASSIGNAR R58 H26
     24: ASSIGNAR R31 H26
     25: ASSIGNAR R34 H26
     26: ASSIGNAR R55 H26
     27: ASSIGNAR R7  H26
     28: ASSIGNAR R49 H26
     29: ASSIGNAR R57 H24
     30: ASSIGNAR R13 H24
     31: ASSIGNAR R12 H24
     32: ASSIGNAR R42 H24
     33: ASSIGNAR R35 H24
     34: ASSIGNAR R25 H24
     35: ASSIGNAR R40 H24
     36: ASSIGNAR R27 H24
     37: ASSIGNAR R47 H24
     38: ASSIGNAR R26 H23
     39: ASSIGNAR R19 H23
     40: ASSIGNAR R50 H23
     41: ASSIGNAR R51 H23
     42: ASSIGNAR R22 H22
     43: ASSIGNAR R33 H21
     44: ASSIGNAR R28 H19
     45: ASSIGNAR R36 H19
     46: ASSIGNAR R53 H19
     47: ASSIGNAR R32 H19
```

```
48: ASSIGNAR R11 H19
49: ASSIGNAR R38 H17
50: ASSIGNAR R9 H17
51: ASSIGNAR R39 H17
52: ASSIGNAR R60 H17
53: ASSIGNAR R46 H16
54: ASSIGNAR R59 H16
55: ASSIGNAR R48 H15
56: ASSIGNAR R10 H15
57: ASSIGNAR R17 H14
58: ASSIGNAR R52 H10
59: ASSIGNAR R56 H9
```

```
time spent:  0.02 seconds instantiating 0 easy, 1263 hard action templates
            0.00 seconds reachability analysis, yielding 3171 facts and 1263
actions
            0.01 seconds creating final representation with 3159 relevant facts, 0
relevant fluents
            0.03 seconds computing LNF
            0.00 seconds building connectivity graph
            3.38 seconds searching, evaluating 31000 states, to a max depth of 0
            3.44 seconds total time
```

Mida 70

Shell

ff: found legal plan as follows

```
step  0: ASSIGNAR R8 H35
      1: ASSIGNAR R30 H35
      2: ASSIGNAR R18 H35
      3: ASSIGNAR R32 H35
      4: ASSIGNAR R57 H35
      5: ASSIGNAR R15 H35
      6: ASSIGNAR R64 H35
      7: ASSIGNAR R9 H35
      8: ASSIGNAR R38 H35
      9: ASSIGNAR R61 H35
     10: ASSIGNAR R10 H35
     11: ASSIGNAR R42 H35
     12: ASSIGNAR R16 H34
     13: ASSIGNAR R24 H34
     14: ASSIGNAR R17 H34
     15: ASSIGNAR R1 H34
     16: ASSIGNAR R68 H34
     17: ASSIGNAR R27 H34
     18: ASSIGNAR R4 H34
     19: ASSIGNAR R6 H34
     20: ASSIGNAR R44 H34
     21: ASSIGNAR R62 H34
```

22: ASSIGNAR R19 H33
23: ASSIGNAR R26 H33
24: ASSIGNAR R3 H33
25: ASSIGNAR R12 H33
26: ASSIGNAR R21 H33
27: ASSIGNAR R59 H33
28: ASSIGNAR R51 H33
29: ASSIGNAR R45 H33
30: ASSIGNAR R69 H33
31: ASSIGNAR R70 H32
32: ASSIGNAR R49 H32
33: ASSIGNAR R67 H32
34: ASSIGNAR R60 H32
35: ASSIGNAR R28 H32
36: ASSIGNAR R65 H32
37: ASSIGNAR R31 H31
38: ASSIGNAR R50 H31
39: ASSIGNAR R2 H31
40: ASSIGNAR R52 H30
41: ASSIGNAR R25 H29
42: ASSIGNAR R35 H29
43: ASSIGNAR R7 H29
44: ASSIGNAR R43 H29
45: ASSIGNAR R14 H29
46: ASSIGNAR R11 H29
47: ASSIGNAR R33 H28
48: ASSIGNAR R34 H28
49: ASSIGNAR R5 H28
50: ASSIGNAR R37 H25
51: ASSIGNAR R66 H25
52: ASSIGNAR R41 H25
53: ASSIGNAR R63 H25
54: ASSIGNAR R13 H25
55: ASSIGNAR R58 H25
56: ASSIGNAR R20 H25
57: ASSIGNAR R48 H25
58: ASSIGNAR R55 H24
59: ASSIGNAR R22 H24
60: ASSIGNAR R39 H24
61: ASSIGNAR R23 H24
62: ASSIGNAR R36 H23
63: ASSIGNAR R47 H23
64: ASSIGNAR R40 H23
65: ASSIGNAR R46 H22
66: ASSIGNAR R53 H22
67: ASSIGNAR R29 H19
68: ASSIGNAR R56 H11
69: ASSIGNAR R54 H11

time spent: 0.03 seconds instantiating 0 easy, 1642 hard action templates
0.00 seconds reachability analysis, yielding 3834 facts and 1642
actions

```
0.01 seconds creating final representation with 3786 relevant facts, 0
relevant fluents
0.03 seconds computing LNF
0.01 seconds building connectivity graph
6.48 seconds searching, evaluating 49958 states, to a max depth of 0
6.56 seconds total time
```

Extensió 1

Shell

```
python3 ./generate_problems.py 3 --level ext1
./Metric-FF/ff -0 -o ./domain-ext1.pddl -f ./jocs-prova/problem-ext1-auto-3.pddl
```

Mida 3

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R3 H1
         1: ASSIGNAR R2 H1
         2: ASSIGNAR R1 H1
         3: REACH-GOAL
```

```
time spent: 0.00 seconds instantiating 3 easy, 5 hard action templates
            0.00 seconds reachability analysis, yielding 91 facts and 16 actions
            0.00 seconds creating final representation with 46 relevant facts, 1
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.00 seconds searching, evaluating 19 states, to a max depth of 0
            0.00 seconds total time
```

Mida 5

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R5 H1
         1: ASSIGNAR R4 H1
         2: ASSIGNAR R3 H1
         3: ASSIGNAR R2 H1
         4: ASSIGNAR R1 H2
```

5: REACH-GOAL

```
time spent:    0.00 seconds instantiating 5 easy, 11 hard action templates
              0.00 seconds reachability analysis, yielding 141 facts and 48 actions
              0.00 seconds creating final representation with 72 relevant facts, 1
relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.00 seconds searching, evaluating 49 states, to a max depth of 0
              0.00 seconds total time
```

Mida 7

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R7 H1
        1: ASSIGNAR R6 H1
        2: ASSIGNAR R5 H2
        3: ASSIGNAR R4 H2
        4: ASSIGNAR R3 H1
        5: ASSIGNAR R2 H1
        6: ASSIGNAR R1 H1
        7: REACH-GOAL
```

```
time spent:    0.00 seconds instantiating 7 easy, 21 hard action templates
              0.00 seconds reachability analysis, yielding 232 facts and 156 actions
              0.00 seconds creating final representation with 176 relevant facts, 1
relevant fluents
              0.00 seconds computing LNF
              0.00 seconds building connectivity graph
              0.01 seconds searching, evaluating 110 states, to a max depth of 0
              0.01 seconds total time
```

Mida 9

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R9 H1
        1: ASSIGNAR R8 H1
        2: ASSIGNAR R7 H2
        3: ASSIGNAR R6 H3
        4: ASSIGNAR R5 H1
        5: ASSIGNAR R4 H1
```

```
6: ASSIGNAR R3 H1
7: ASSIGNAR R2 H3
8: REBUTJAR R1
9: REACH-GOAL
```

```
time spent:  0.00 seconds instantiating 9 easy, 32 hard action templates
            0.00 seconds reachability analysis, yielding 274 facts and 553 actions
            0.01 seconds creating final representation with 181 relevant facts, 1
relevant fluents
            0.00 seconds computing LNF
            0.01 seconds building connectivity graph
            0.01 seconds searching, evaluating 187 states, to a max depth of 0
            0.03 seconds total time
```

Mida 10

Shell

ff: found legal plan as follows

```
step  0: ASSIGNAR R10 H1
      1: ASSIGNAR R9 H1
      2: ASSIGNAR R8 H1
      3: ASSIGNAR R7 H2
      4: ASSIGNAR R6 H1
      5: ASSIGNAR R5 H1
      6: ASSIGNAR R4 H1
      7: ASSIGNAR R3 H2
      8: ASSIGNAR R2 H1
      9: ASSIGNAR R1 H1
     10: REACH-GOAL
```

```
time spent:  0.01 seconds instantiating 10 easy, 36 hard action templates
            0.00 seconds reachability analysis, yielding 285 facts and 1070 actions
            0.01 seconds creating final representation with 195 relevant facts, 1
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.03 seconds searching, evaluating 252 states, to a max depth of 0
            0.05 seconds total time
```

Mida 11

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R11 H1
        1: ASSIGNAR R10 H1
        2: ASSIGNAR R9 H1
        3: ASSIGNAR R8 H1
        4: ASSIGNAR R7 H2
        5: ASSIGNAR R6 H1
        6: ASSIGNAR R5 H1
        7: ASSIGNAR R4 H3
        8: ASSIGNAR R3 H1
        9: ASSIGNAR R2 H1
       10: ASSIGNAR R1 H1
       11: REACH-GOAL

time spent:  0.04 seconds instantiating 11 easy, 59 hard action templates
            0.00 seconds reachability analysis, yielding 363 facts and 2118 actions
            0.04 seconds creating final representation with 264 relevant facts, 1
relevant fluents
            0.01 seconds computing LNF
            0.00 seconds building connectivity graph
            0.04 seconds searching, evaluating 418 states, to a max depth of 0
            0.13 seconds total time
```

Mida 12

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R12 H1
        1: ASSIGNAR R11 H1
        2: ASSIGNAR R10 H1
        3: ASSIGNAR R9 H3
        4: ASSIGNAR R8 H1
        5: ASSIGNAR R7 H1
        6: ASSIGNAR R6 H1
        7: ASSIGNAR R5 H3
        8: ASSIGNAR R4 H3
        9: ASSIGNAR R3 H5
       10: ASSIGNAR R2 H2
       11: ASSIGNAR R1 H2
       12: REACH-GOAL

time spent:  0.15 seconds instantiating 12 easy, 49 hard action templates
            0.00 seconds reachability analysis, yielding 366 facts and 4157 actions
            0.10 seconds creating final representation with 276 relevant facts, 1
relevant fluents
            0.02 seconds computing LNF
            0.00 seconds building connectivity graph
            0.16 seconds searching, evaluating 388 states, to a max depth of 0
```

0.43 seconds total time

Mida 13

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R13 H1
        1: ASSIGNAR R12 H1
        2: ASSIGNAR R11 H3
        3: ASSIGNAR R10 H1
        4: ASSIGNAR R9 H4
        5: ASSIGNAR R8 H2
        6: ASSIGNAR R7 H3
        7: ASSIGNAR R6 H1
        8: ASSIGNAR R5 H3
        9: ASSIGNAR R4 H1
       10: ASSIGNAR R3 H1
       11: ASSIGNAR R2 H4
       12: REBUTJAR R1
       13: REACH-GOAL
```

time spent: 0.94 seconds instantiating 13 easy, 72 hard action templates
0.00 seconds reachability analysis, yielding 473 facts and 8277 actions
1.00 seconds creating final representation with 403 relevant facts, 1

relevant fluents

0.04 seconds computing LNF
0.03 seconds building connectivity graph
0.33 seconds searching, evaluating 528 states, to a max depth of 0
2.34 seconds total time

Mida 14

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R14 H1
        1: ASSIGNAR R13 H1
        2: ASSIGNAR R12 H1
        3: ASSIGNAR R11 H1
        4: ASSIGNAR R10 H1
        5: ASSIGNAR R8 H2
        6: ASSIGNAR R7 H3
        7: ASSIGNAR R6 H2
        8: ASSIGNAR R5 H4
        9: ASSIGNAR R4 H2
```



```
10: ASSIGNAR R3 H1
11: ASSIGNAR R2 H2
12: ASSIGNAR R1 H3
13: REBUTJAR R9
14: REACH-GOAL
```

```
time spent: 6.13 seconds instantiating 14 easy, 58 hard action templates
            0.00 seconds reachability analysis, yielding 434 facts and 16456
actions
            5.46 seconds creating final representation with 335 relevant facts, 1
relevant fluents
            0.07 seconds computing LNF
            0.02 seconds building connectivity graph
            0.65 seconds searching, evaluating 453 states, to a max depth of 0
            12.33 seconds total time
```

Mida 15

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R15 H1
      1: ASSIGNAR R14 H3
      2: ASSIGNAR R13 H3
      3: ASSIGNAR R12 H1
      4: ASSIGNAR R11 H1
      5: ASSIGNAR R10 H1
      6: ASSIGNAR R9 H2
      7: ASSIGNAR R8 H6
      8: ASSIGNAR R7 H1
      9: ASSIGNAR R6 H1
     10: ASSIGNAR R5 H1
     11: ASSIGNAR R4 H2
     12: ASSIGNAR R3 H3
     13: ASSIGNAR R2 H2
     14: ASSIGNAR R1 H2
     15: REACH-GOAL
```

```
time spent: 34.67 seconds instantiating 15 easy, 94 hard action templates
            0.00 seconds reachability analysis, yielding 567 facts and 32877
actions
            29.14 seconds creating final representation with 501 relevant facts, 1
relevant fluents
            0.16 seconds computing LNF
            0.08 seconds building connectivity graph
            3.56 seconds searching, evaluating 889 states, to a max depth of 0
            67.61 seconds total time
```


Extensió 3

Shell

```
python3 ./generate_problems.py 3 --level ext3
./Metric-FF/ff -0 -o ./domain-ext3.pddl -f ./jocs-prova/problem-ext3-auto-3.pddl
```

Mida 3

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R3 H1
         1: ASSIGNAR R1 H1
         2: ASSIGNAR R2 H2
         3: REACH-GOAL
```

```
time spent:  0.00 seconds instantiating 3 easy, 6 hard action templates
            0.00 seconds reachability analysis, yielding 90 facts and 17 actions
            0.00 seconds creating final representation with 43 relevant facts, 1
```

relevant fluents

```
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.00 seconds searching, evaluating 19 states, to a max depth of 0
            0.00 seconds total time
```

Mida 5

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R5 H1
         1: ASSIGNAR R3 H1
         2: ASSIGNAR R2 H1
         3: ASSIGNAR R1 H3
         4: ASSIGNAR R4 H2
         5: REACH-GOAL
```

```
time spent:  0.01 seconds instantiating 5 easy, 11 hard action templates
            0.00 seconds reachability analysis, yielding 158 facts and 48 actions
            0.00 seconds creating final representation with 106 relevant facts, 1
```

relevant fluents

```
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.00 seconds searching, evaluating 47 states, to a max depth of 0
            0.01 seconds total time
```

Mida 7

Shell

ff: found legal plan as follows

step 0: ASSIGNAR R7 H3
 1: ASSIGNAR R6 H4
 2: ASSIGNAR R5 H1
 3: ASSIGNAR R4 H1
 4: ASSIGNAR R3 H3
 5: ASSIGNAR R1 H2
 6: REBUTJAR R2
 7: REACH-GOAL

time spent: 0.00 seconds instantiating 7 easy, 19 hard action templates
 0.00 seconds reachability analysis, yielding 194 facts and 154 actions
 0.00 seconds creating final representation with 102 relevant facts, 1

relevant fluents

 0.01 seconds computing LNF
 0.00 seconds building connectivity graph
 0.00 seconds searching, evaluating 88 states, to a max depth of 0
 0.01 seconds total time

Mida 9

Shell

ff: found legal plan as follows

step 0: ASSIGNAR R9 H2
 1: ASSIGNAR R8 H2
 2: ASSIGNAR R7 H5
 3: ASSIGNAR R6 H3
 4: ASSIGNAR R5 H2
 5: ASSIGNAR R4 H3
 6: ASSIGNAR R3 H4
 7: ASSIGNAR R2 H1
 8: ASSIGNAR R1 H5
 9: REACH-GOAL

time spent: 0.01 seconds instantiating 9 easy, 31 hard action templates
 0.00 seconds reachability analysis, yielding 281 facts and 552 actions
 0.00 seconds creating final representation with 196 relevant facts, 1

relevant fluents

 0.00 seconds computing LNF
 0.00 seconds building connectivity graph
 0.02 seconds searching, evaluating 173 states, to a max depth of 0
 0.03 seconds total time

Mida 10

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R10 H2
        1: ASSIGNAR R8  H2
        2: ASSIGNAR R6  H1
        3: ASSIGNAR R9  H2
        4: ASSIGNAR R5  H2
        5: ASSIGNAR R1  H3
        6: ASSIGNAR R7  H2
        7: ASSIGNAR R4  H1
        8: ASSIGNAR R3  H1
        9: ASSIGNAR R2  H1
       10: REACH-GOAL
```

```
time spent:  0.01 seconds instantiating 10 easy, 49 hard action templates
            0.00 seconds reachability analysis, yielding 329 facts and 1083 actions
            0.00 seconds creating final representation with 270 relevant facts, 1
```

relevant fluents

```
0.01 seconds computing LNF
0.00 seconds building connectivity graph
0.00 seconds searching, evaluating 304 states, to a max depth of 0
0.02 seconds total time
```

Mida 11

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R10 H1
        1: ASSIGNAR R8  H2
        2: ASSIGNAR R6  H5
        3: ASSIGNAR R4  H5
        4: ASSIGNAR R3  H4
        5: ASSIGNAR R2  H1
        6: ASSIGNAR R11 H2
        7: ASSIGNAR R9  H3
        8: ASSIGNAR R7  H2
        9: ASSIGNAR R5  H6
       10: ASSIGNAR R1  H3
       11: REACH-GOAL
```

```
time spent:  0.03 seconds instantiating 11 easy, 48 hard action templates
            0.00 seconds reachability analysis, yielding 369 facts and 2107 actions
            0.01 seconds creating final representation with 287 relevant facts, 1
```

relevant fluents

```
0.02 seconds computing LNF
0.00 seconds building connectivity graph
```

```
0.01 seconds searching, evaluating 354 states, to a max depth of 0
0.07 seconds total time
```

Mida 12

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R10 H3
         1: ASSIGNAR R9 H3
         2: ASSIGNAR R8 H2
         3: ASSIGNAR R6 H1
         4: ASSIGNAR R3 H1
         5: ASSIGNAR R2 H1
         6: ASSIGNAR R12 H3
         7: ASSIGNAR R11 H3
         8: ASSIGNAR R7 H3
         9: ASSIGNAR R5 H4
        10: ASSIGNAR R4 H1
        11: ASSIGNAR R1 H5
        12: REACH-GOAL
```

```
time spent:  0.06 seconds instantiating 12 easy, 47 hard action templates
            0.00 seconds reachability analysis, yielding 369 facts and 4155 actions
            0.04 seconds creating final representation with 284 relevant facts, 1
```

relevant fluents

```
0.02 seconds computing LNF
0.00 seconds building connectivity graph
0.05 seconds searching, evaluating 343 states, to a max depth of 0
0.17 seconds total time
```

Mida 13

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R13 H2
         1: ASSIGNAR R12 H4
         2: ASSIGNAR R10 H3
         3: ASSIGNAR R9 H4
         4: ASSIGNAR R8 H2
         5: ASSIGNAR R4 H3
         6: ASSIGNAR R3 H2
         7: ASSIGNAR R2 H1
         8: ASSIGNAR R1 H3
         9: ASSIGNAR R11 H3
        10: ASSIGNAR R7 H3
```

```
11: ASSIGNAR R6 H7
12: ASSIGNAR R5 H7
13: REACH-GOAL
```

```
time spent: 0.41 seconds instantiating 13 easy, 76 hard action templates
            0.00 seconds reachability analysis, yielding 485 facts and 8281 actions
            0.43 seconds creating final representation with 423 relevant facts, 1
relevant fluents
            0.03 seconds computing LNF
            0.01 seconds building connectivity graph
            0.16 seconds searching, evaluating 563 states, to a max depth of 0
            1.04 seconds total time
```

Mida 14

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R14 H1
      1: ASSIGNAR R13 H2
      2: ASSIGNAR R12 H2
      3: ASSIGNAR R11 H4
      4: ASSIGNAR R10 H6
      5: ASSIGNAR R9 H4
      6: ASSIGNAR R8 H2
      7: ASSIGNAR R7 H6
      8: ASSIGNAR R6 H2
      9: ASSIGNAR R5 H1
     10: ASSIGNAR R4 H7
     11: ASSIGNAR R3 H6
     12: ASSIGNAR R2 H3
     13: ASSIGNAR R1 H3
     14: REACH-GOAL
```

```
time spent: 1.75 seconds instantiating 14 easy, 66 hard action templates
            0.00 seconds reachability analysis, yielding 442 facts and 16464
actions
            1.62 seconds creating final representation with 343 relevant facts, 1
relevant fluents
            0.04 seconds computing LNF
            0.01 seconds building connectivity graph
            0.69 seconds searching, evaluating 603 states, to a max depth of 0
            4.11 seconds total time
```

Mida 15

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R14 H2
        1: ASSIGNAR R13 H8
        2: ASSIGNAR R12 H3
        3: ASSIGNAR R11 H2
        4: ASSIGNAR R10 H3
        5: ASSIGNAR R9 H3
        6: ASSIGNAR R7 H3
        7: ASSIGNAR R6 H3
        8: ASSIGNAR R5 H3
        9: ASSIGNAR R4 H1
       10: ASSIGNAR R3 H4
       11: ASSIGNAR R2 H4
       12: ASSIGNAR R1 H2
       13: ASSIGNAR R15 H8
       14: ASSIGNAR R8 H2
       15: REACH-GOAL
```

```
time spent:  9.93 seconds instantiating 15 easy, 90 hard action templates
            0.00 seconds reachability analysis, yielding 538 facts and 32873
```

actions

```
            7.92 seconds creating final representation with 447 relevant facts, 1
relevant fluents
```

```
            0.11 seconds computing LNF
            0.05 seconds building connectivity graph
            2.00 seconds searching, evaluating 771 states, to a max depth of 0
            20.01 seconds total time
```


Extensió 4

Shell

```
python3 ./generate_problems.py 3 --level ext4
./Metric-FF/ff -O -o ./domain-ext4.pddl -f ./jocs-prova/problem-ext1-auto-4.pddl
```

Mida 3

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R3 H1
         1: ASSIGNAR R1 H1
         2: ASSIGNAR R2 H1
         3: REACH-GOAL
```

```
time spent:  0.00 seconds instantiating 3 easy, 4 hard action templates
             0.00 seconds reachability analysis, yielding 126 facts and 15 actions
             0.00 seconds creating final representation with 47 relevant facts, 1
```

relevant fluents

```
             0.00 seconds computing LNF
             0.00 seconds building connectivity graph
             0.00 seconds searching, evaluating 17 states, to a max depth of 0
             0.00 seconds total time
```

Mida 5

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R5 H1
         1: ASSIGNAR R1 H1
         2: ASSIGNAR R3 H1
         3: ASSIGNAR R4 H1
         4: ASSIGNAR R2 H2
         5: REACH-GOAL
```

```
time spent:  0.01 seconds instantiating 5 easy, 14 hard action templates
             0.00 seconds reachability analysis, yielding 189 facts and 51 actions
             0.00 seconds creating final representation with 89 relevant facts, 1
```

relevant fluents

```
             0.00 seconds computing LNF
             0.00 seconds building connectivity graph
             0.00 seconds searching, evaluating 57 states, to a max depth of 0
             0.01 seconds total time
```

Mida 7

Shell

ff: found legal plan as follows

step 0: ASSIGNAR R7 H1
 1: ASSIGNAR R4 H1
 2: ASSIGNAR R2 H1
 3: ASSIGNAR R5 H1
 4: ASSIGNAR R3 H1
 5: ASSIGNAR R6 H2
 6: ASSIGNAR R1 H2
 7: REACH-GOAL

time spent: 0.01 seconds instantiating 7 easy, 22 hard action templates
 0.00 seconds reachability analysis, yielding 263 facts and 157 actions
 0.00 seconds creating final representation with 155 relevant facts, 1

relevant fluents

 0.00 seconds computing LNF
 0.00 seconds building connectivity graph
 0.00 seconds searching, evaluating 121 states, to a max depth of 0
 0.01 seconds total time

Mida 9

Shell

ff: found legal plan as follows

step 0: ASSIGNAR R9 H1
 1: ASSIGNAR R4 H1
 2: ASSIGNAR R3 H1
 3: ASSIGNAR R2 H1
 4: ASSIGNAR R6 H1
 5: ASSIGNAR R7 H5
 6: ASSIGNAR R5 H5
 7: REBUTJAR R8
 8: REBUTJAR R1
 9: REACH-GOAL

time spent: 0.01 seconds instantiating 9 easy, 19 hard action templates
 0.00 seconds reachability analysis, yielding 298 facts and 540 actions
 0.00 seconds creating final representation with 154 relevant facts, 1

relevant fluents

 0.00 seconds computing LNF
 0.00 seconds building connectivity graph
 0.02 seconds searching, evaluating 120 states, to a max depth of 0
 0.03 seconds total time

Mida 10

Shell

ff: found legal plan as follows

```
step  0: ASSIGNAR R10 H1
      1: ASSIGNAR R8  H1
      2: ASSIGNAR R6  H1
      3: ASSIGNAR R2  H1
      4: ASSIGNAR R4  H1
      5: ASSIGNAR R3  H1
      6: ASSIGNAR R9  H2
      7: ASSIGNAR R5  H4
      8: REBUTJAR R7
      9: REBUTJAR R1
     10: REACH-GOAL
```

```
time spent:  0.00 seconds instantiating 10 easy, 24 hard action templates
            0.00 seconds reachability analysis, yielding 300 facts and 1058 actions
            0.01 seconds creating final representation with 147 relevant facts, 1
relevant fluents
            0.00 seconds computing LNF
            0.00 seconds building connectivity graph
            0.02 seconds searching, evaluating 178 states, to a max depth of 0
            0.03 seconds total time
```

Mida 11

Shell

ff: found legal plan as follows

```
step  0: ASSIGNAR R11 H1
      1: ASSIGNAR R9  H1
      2: ASSIGNAR R7  H1
      3: ASSIGNAR R6  H1
      4: ASSIGNAR R5  H1
      5: ASSIGNAR R3  H1
      6: ASSIGNAR R2  H1
      7: ASSIGNAR R8  H3
      8: ASSIGNAR R4  H3
      9: ASSIGNAR R10 H3
     10: ASSIGNAR R1  H2
     11: REACH-GOAL
```

```
time spent:  0.01 seconds instantiating 11 easy, 49 hard action templates
            0.00 seconds reachability analysis, yielding 444 facts and 2108 actions
```

```
0.02 seconds creating final representation with 342 relevant facts, 1
relevant fluents
0.00 seconds computing LNF
0.01 seconds building connectivity graph
0.02 seconds searching, evaluating 345 states, to a max depth of 0
0.06 seconds total time
```

Mida 12

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R12 H1
      1: ASSIGNAR R10 H1
      2: ASSIGNAR R7 H1
      3: ASSIGNAR R5 H1
      4: ASSIGNAR R4 H1
      5: ASSIGNAR R3 H1
      6: ASSIGNAR R2 H1
      7: ASSIGNAR R11 H2
      8: ASSIGNAR R8 H2
      9: ASSIGNAR R9 H3
     10: ASSIGNAR R6 H3
     11: ASSIGNAR R1 H3
     12: REACH-GOAL
```

```
time spent: 0.03 seconds instantiating 12 easy, 36 hard action templates
            0.00 seconds reachability analysis, yielding 404 facts and 4144 actions
            0.06 seconds creating final representation with 269 relevant facts, 1
relevant fluents
            0.01 seconds computing LNF
            0.00 seconds building connectivity graph
            0.02 seconds searching, evaluating 306 states, to a max depth of 0
            0.12 seconds total time
```

Mida 13

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R13 H2
      1: ASSIGNAR R10 H2
      2: ASSIGNAR R3 H2
      3: ASSIGNAR R11 H2
      4: ASSIGNAR R1 H2
      5: ASSIGNAR R12 H3
      6: ASSIGNAR R5 H3
```

```
7: ASSIGNAR R4 H3
8: ASSIGNAR R9 H1
9: ASSIGNAR R7 H1
10: ASSIGNAR R8 H1
11: ASSIGNAR R2 H1
12: ASSIGNAR R6 H4
13: REACH-GOAL
```

```
time spent: 0.46 seconds instantiating 13 easy, 65 hard action templates
            0.00 seconds reachability analysis, yielding 532 facts and 8270 actions
            0.28 seconds creating final representation with 428 relevant facts, 1
relevant fluents
            0.03 seconds computing LNF
            0.00 seconds building connectivity graph
            0.16 seconds searching, evaluating 474 states, to a max depth of 0
            0.93 seconds total time
```

Mida 14

Shell

ff: found legal plan as follows

```
step 0: ASSIGNAR R14 H1
      1: ASSIGNAR R13 H1
      2: ASSIGNAR R6 H1
      3: ASSIGNAR R1 H1
      4: ASSIGNAR R12 H1
      5: ASSIGNAR R10 H1
      6: ASSIGNAR R9 H1
      7: ASSIGNAR R2 H1
      8: ASSIGNAR R11 H3
      9: ASSIGNAR R8 H3
     10: ASSIGNAR R7 H3
     11: ASSIGNAR R4 H3
     12: ASSIGNAR R5 H2
     13: ASSIGNAR R3 H7
     14: REACH-GOAL
```

```
time spent: 1.63 seconds instantiating 14 easy, 73 hard action templates
            0.00 seconds reachability analysis, yielding 546 facts and 16471
actions
            1.76 seconds creating final representation with 442 relevant facts, 1
relevant fluents
            0.06 seconds computing LNF
            0.01 seconds building connectivity graph
            0.71 seconds searching, evaluating 670 states, to a max depth of 0
            4.17 seconds total time
```

Mida 15

Shell

ff: found legal plan as follows

```
step    0: ASSIGNAR R15 H1
        1: ASSIGNAR R11 H1
        2: ASSIGNAR R9 H1
        3: ASSIGNAR R2 H1
        4: ASSIGNAR R14 H1
        5: ASSIGNAR R8 H1
        6: ASSIGNAR R13 H1
        7: ASSIGNAR R12 H1
        8: ASSIGNAR R6 H1
        9: ASSIGNAR R1 H1
       10: ASSIGNAR R10 H6
       11: ASSIGNAR R7 H6
       12: ASSIGNAR R3 H6
       13: ASSIGNAR R5 H3
       14: ASSIGNAR R4 H2
       15: REACH-GOAL
```

```
time spent:    8.87 seconds instantiating 15 easy, 83 hard action templates
              0.00 seconds reachability analysis, yielding 591 facts and 32866
```

actions

```
              10.73 seconds creating final representation with 454 relevant facts, 1
relevant fluents
              0.16 seconds computing LNF
              0.11 seconds building connectivity graph
              4.57 seconds searching, evaluating 751 states, to a max depth of 0
              24.44 seconds total time
```